

Random Forest Classification

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(tidyverse)
library(rio)
library(ggplot2)
library(psych)
library(ggthemes)
library(summarytools)
library(tidyr)
library(formatR)
library(sjmisc)
library(caret)
library(randomForest)
library(report)
```

load inbuilt data

```
data(iris)
head(iris)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

data structure

```
str(iris)
```

```
'data.frame':  150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1
1 1 1 1 1 1 ...
```

view data set

```
View(iris)
```

SET SEED: to ensure reproducibility

```
set.seed(123)
```

confirm missing values

```
sum(is.na(iris))
```

```
[1] 0
```

```
colSums(is.na(iris))
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	0	0	0	0

split data set

train model

```
train_iris <- sample(seq_len(nrow(iris)), size = 0.7 * nrow(iris))
train_iris_data <- iris[train_iris, ]
test_iris <- iris[-train_iris, ]
```

check dimension

```
dim(train_iris_data)
```

```
[1] 105  5
```

```
dim(test_iris)
```

```
[1] 45  5
```

head

```
head(train_iris_data)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
14	4.3	3.0	1.1	0.1	setosa
50	5.0	3.3	1.4	0.2	setosa
118	7.7	3.8	6.7	2.2	virginica
43	4.4	3.2	1.3	0.2	setosa
150	5.9	3.0	5.1	1.8	virginica
148	6.5	3.0	5.2	2.0	virginica

train random forest classify model

```
rf_model <- randomForest(formula = Species ~ ., data = train_iris_data,  
  ntree = 500,  
  mtry = 2, importance = TRUE, proximity = FALSE)
```

print model summary

```
print(rf_model)
```

```
Call:
randomForest(formula = Species ~ ., data = train_iris_data, ntree = 500,
             mtry = 2, importance = TRUE, proximity = FALSE)
Type of random forest: classification
Number of trees: 500
No. of variables tried at each split: 2

OOB estimate of error rate: 5.71%
Confusion matrix:
      setosa versicolor virginica class.error
setosa      36          0          0 0.00000000
versicolor   0         29          3 0.09375000
virginica    0          3         34 0.08108108
```

predict on test set

```
predictions <- predict(rf_model, newdata = test_iris)
head(predictions)
```

```
      1      2      3      5     11     18
setosa setosa setosa setosa setosa setosa
Levels: setosa versicolor virginica
```

```
head(test_iris)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
11	5.4	3.7	1.5	0.2	setosa
18	5.1	3.5	1.4	0.3	setosa

model evaluation performanve

```
conf_mat <- table(pridicted = predictions, actual = test_iris$Species)
print(conf_mat)
```

	actual		
pridicted	setosa	versicolor	virginica
setosa	14	0	0
versicolor	0	17	0
virginica	0	1	13

```
accuracy <- sum(diag(conf_mat))/sum(conf_mat)
cat("test Accuracy:", round(accuracy, 4), "\n")
```

```
test Accuracy: 0.9778
```

variable importance

```
cat(" \n variable importance(mean decrease in accuracy:\n")
```

```
variable importance(mean decrease in accuracy:
```

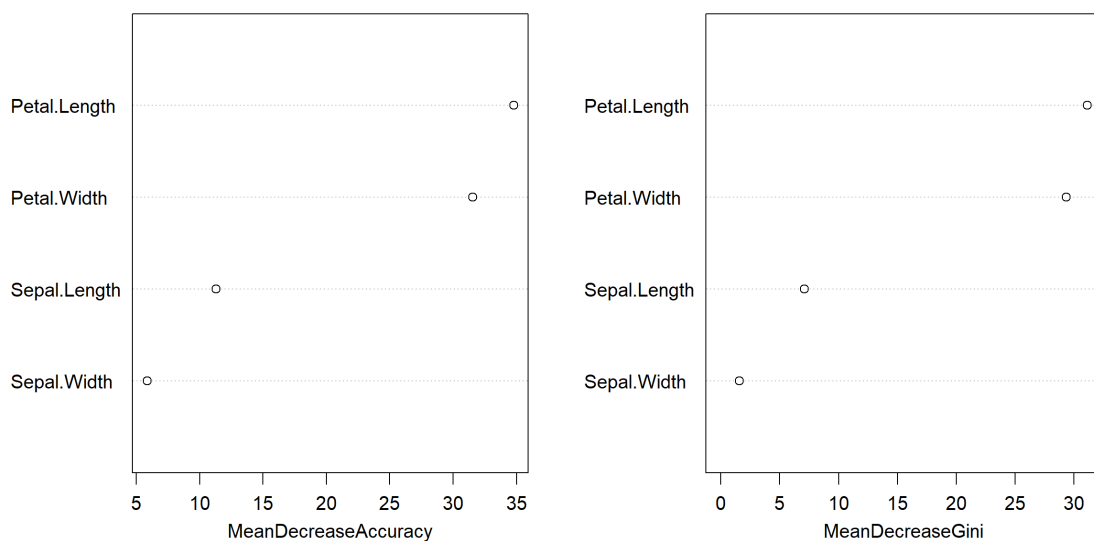
```
print(importance(rf_model, type = 1))
```

	MeanDecreaseAccuracy
Sepal.Length	11.294701
Sepal.Width	5.875234
Petal.Length	34.780877
Petal.Width	31.534277

variable importance

```
varImpPlot(rf_model, main = " Random Forest variable importance")
```

Random Forest variable importance



creating new data

```
data = data.frame(Sepal.Length = c(4.6, 4.1), Sepal.Width = c(2.6, 3.5),  
  Petal.Length = c(1.3, 1.4), Petal.Width = c(0.3, 0.4))  
head(data)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	4.6	2.6	1.3	0.3
2	4.1	3.5	1.4	0.4

predictioion on new data set

```
predicted <- predict(rf_model, newdata = data)  
head(predicted)
```

1	2
setosa	setosa

Levels: setosa versicolor virginica